

HADAMARD GATES AND QUANTUM INTERFERENCE

Coherent amplitudes, relative phase, basis changes,
interferometers, Walsh-Hadamard transforms, phase oracles,
algorithms, QKD, and noisy experiments

A detailed reference for quantum computing, quantum information, quantum
algorithms, and quantum cryptography

Topic 12 in the Quantum Mathematics Foundation series

Prepared as a self-contained study and revision volume

How to use this volume

The Hadamard gate is often introduced as a device that “creates superposition.” That statement is incomplete. The same gate can create a superposition, erase one, change measurement basis, convert phase information into measurable populations, and cause constructive or destructive interference. Its real significance is that it controls how probability *amplitudes* are recombined.

This chapter is designed to be read after the foundations of complex amplitudes, vectors, matrices, inner products, qubits, measurement, and single-qubit gates. It is still self-contained enough to serve as a later reference.

Learning objectives

After completing the volume, you should be able to:

- calculate the action of H on basis states and arbitrary qubits;
- explain interference using coherent sums of complex amplitudes;
- derive the probabilities produced by an H -phase- H interferometer;
- distinguish global phase, relative phase, classical mixing, and coherent superposition;
- use H as a change of basis for states, measurements, and operators;
- derive $H^{\otimes n}$ and the Walsh-Hadamard transform;
- explain how phase oracles become useful only after interference;
- recognize the role of Hadamard interference in Deutsch, Bernstein-Vazirani, Grover, phase estimation, and QKD;
- predict how dephasing and which-path information reduce interference visibility;
- implement and verify the main experiments in Qiskit.

Computational-basis vectors are $|0\rangle = (1, 0)^T$ and $|1\rangle = (0, 1)^T$. The phase-rotation convention is $R_z(\phi) = \exp(-i\phi Z/2)$. When a phase gate $P(\phi) = \text{diag}(1, e^{i\phi})$ is used instead, it differs from $R_z(\phi)$ by a global phase. Measurement probabilities are unchanged.

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1. Interference is the operational heart of quantum computing

A quantum state assigns a complex amplitude to each alternative. If several coherent alternatives lead to the same final outcome, the amplitudes are added first, and only then is the squared magnitude taken. For two paths with amplitudes a and b ,

$$P = |a + b|^2 = |a|^2 + |b|^2 + 2 \operatorname{Re}(a^* b).$$

The final term is the **interference term**. It can be positive, negative, or zero depending on relative phase.

In a classical probabilistic mixture, mutually exclusive alternatives contribute probabilities:

$$P_{\text{classical}} = P(A) + P(B).$$

No phase-sensitive cross term exists. Quantum advantage does not come merely from storing many amplitudes. It comes from arranging their phases so that unwanted outcomes cancel while useful outcomes reinforce.

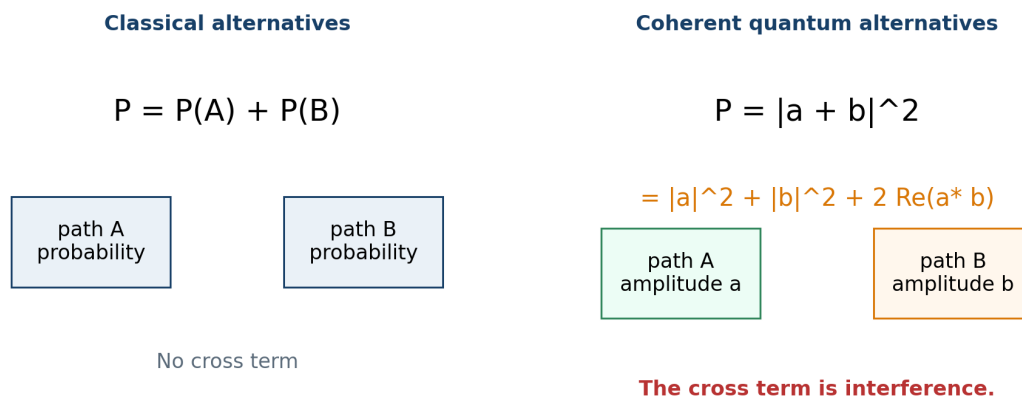


Figure 1: Classical alternatives add probabilities; coherent quantum alternatives add amplitudes and therefore contain an interference cross term.

A useful quantum algorithm has three broad stages: create a coherent set of computational paths, encode problem-dependent information into their amplitudes or phases, and recombine the paths so that measurement favors informative outcomes. Hadamard gates frequently implement the first and third stages.

2. The Hadamard operator

The one-qubit Hadamard gate is

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Its columns are its outputs on the computational basis:

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle, \quad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle.$$

Thus H maps the Z -basis states $\{|0\rangle, |1\rangle\}$ to the X -basis states $\{|+\rangle, |-\rangle\}$.

2.1 Structural properties

$$\begin{aligned}
 H^\dagger &= H && \text{Hermitian,} \\
 H^\dagger H &= I && \text{unitary,} \\
 H^2 &= I && \text{self-inverse,} \\
 \det H &= -1, \\
 \text{Tr}(H) &= 0.
 \end{aligned}$$

Because H is both Hermitian and unitary, its eigenvalues are $+1$ and -1 . It is a reflection in Hilbert space, and, up to a physically irrelevant global phase, it can also be interpreted as a π rotation of the Bloch sphere about the axis $(\hat{x} + \hat{z})/\sqrt{2}$.

A useful Pauli decomposition is

$$H = \frac{X + Z}{\sqrt{2}}.$$

This already suggests why H exchanges bit information (X behavior) and phase information (Z behavior).

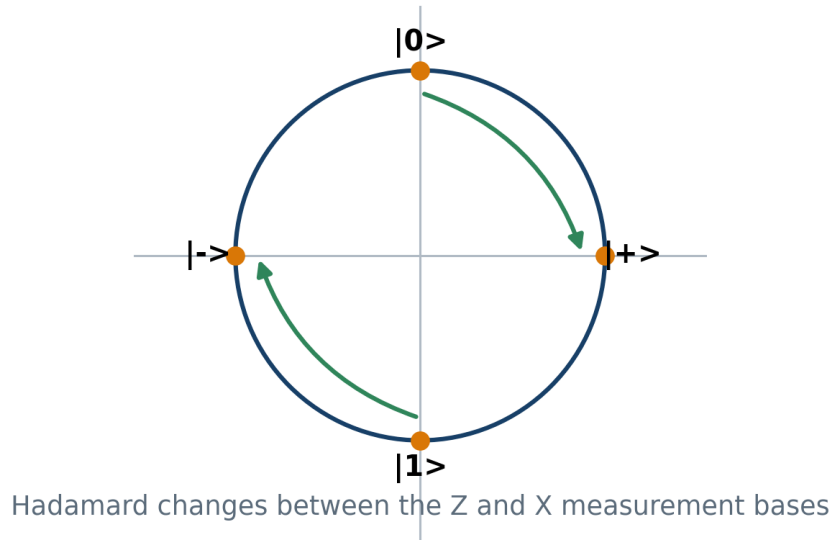


Figure 2: The Hadamard gate changes between the computational (Z) and diagonal (X) bases.

3. Action on an arbitrary qubit

Let

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1.$$

Then

$$\begin{aligned}
 H|\psi\rangle &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \\
 &= \frac{\alpha + \beta}{\sqrt{2}} |0\rangle + \frac{\alpha - \beta}{\sqrt{2}} |1\rangle.
 \end{aligned}$$

The output amplitudes are a **sum** and a **difference**. The corresponding probabilities are

$$P(0) = \frac{|\alpha + \beta|^2}{2} = \frac{|\alpha|^2 + |\beta|^2 + 2\operatorname{Re}(\alpha^*\beta)}{2},$$

$$P(1) = \frac{|\alpha - \beta|^2}{2} = \frac{|\alpha|^2 + |\beta|^2 - 2\operatorname{Re}(\alpha^*\beta)}{2}.$$

Using normalization,

$$P(0) = \frac{1}{2} + \operatorname{Re}(\alpha^*\beta), \quad P(1) = \frac{1}{2} - \operatorname{Re}(\alpha^*\beta).$$

The Hadamard measurement statistics therefore expose the real part of the coherence $\alpha^*\beta$.

Hadamard outputs are coherent sums and differences of input amplitudes

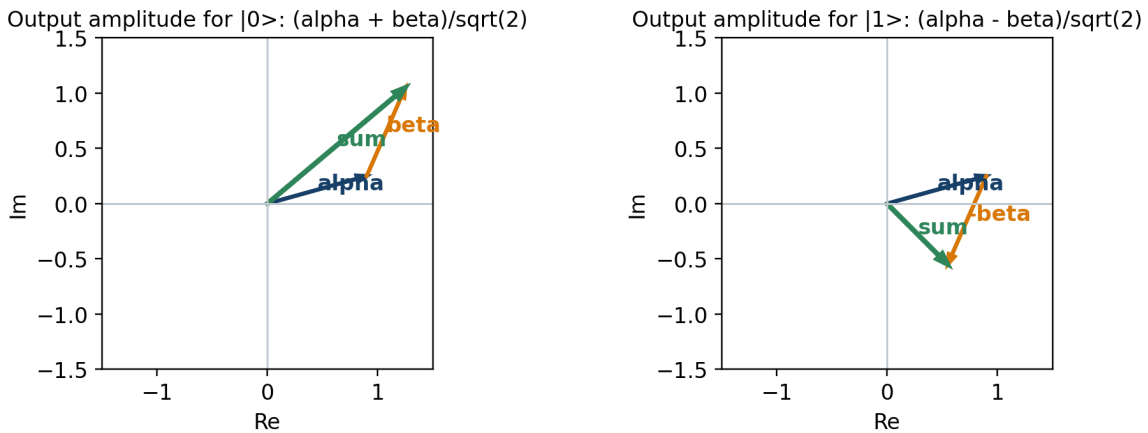


Figure 3: For a general input, the two output amplitudes are coherent vector sums and differences in the complex plane.

Consider

$$|\psi_+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |\psi_-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

Both give $P_Z(0) = P_Z(1) = 1/2$. Yet

$$H|\psi_+\rangle = |0\rangle, \quad H|\psi_-\rangle = |1\rangle.$$

The computational populations before H were identical; their relative phases were different. Interference converts that phase difference into a deterministic classical bit.

4. The Hadamard gate as a basis transformation

A basis is not merely a set of labels. It determines which amplitudes are treated as coordinates and which observable is measured. Since

$$|+\rangle = H|0\rangle, \quad |-\rangle = H|1\rangle,$$

we also have, because $H^{-1} = H$,

$$|0\rangle = H|+\rangle, \quad |1\rangle = H|-\rangle.$$

4.1 Measuring in the X basis

A standard hardware measurement is usually a computational-basis measurement. To measure in the X basis:

1. apply H ;
2. measure in the Z basis.

Indeed,

$$P(+)=|\langle +|\psi\rangle|^2=|\langle 0|H|\psi\rangle|^2,$$

so the probability of outcome $+$ equals the probability of obtaining computational result 0 after applying H .

4.2 Coordinate transformation

If $|\psi\rangle$ has Z -basis coordinate vector $(\alpha, \beta)^T$, its X -basis coordinates are

$$\begin{pmatrix} c_+ \\ c_- \end{pmatrix} = \begin{pmatrix} \langle +|\psi\rangle \\ \langle -|\psi\rangle \end{pmatrix} = H \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

Because $H = H^{-1}$, the same matrix converts in either direction.

State transformation versus coordinate transformation

Applying H physically to a state and merely rewriting the same state in the X basis use the same numerical matrix here, but they are conceptually different operations. In one case the physical state changes; in the other only its coordinate description changes. Always track whether a matrix is acting actively on a state or passively on a basis.

5. Global phase, relative phase, and observable interference

A global phase multiplies the entire state:

$$|\psi\rangle \longrightarrow e^{i\gamma} |\psi\rangle.$$

Every measurement amplitude receives the same factor, so every probability remains unchanged:

$$|\langle m|e^{i\gamma}|\psi\rangle|^2 = |e^{i\gamma}|^2 |\langle m|\psi\rangle|^2 = |\langle m|\psi\rangle|^2.$$

A relative phase changes one component with respect to another:

$$|\psi(\phi)\rangle = \frac{|0\rangle + e^{i\phi}|1\rangle}{\sqrt{2}}.$$

All these states give equal probabilities in the Z basis, but after a Hadamard gate,

$$\begin{aligned} H|\psi(\phi)\rangle &= \frac{1+e^{i\phi}}{2}|0\rangle + \frac{1-e^{i\phi}}{2}|1\rangle, \\ P(0) &= \frac{|1+e^{i\phi}|^2}{4} = \cos^2 \frac{\phi}{2} = \frac{1+\cos \phi}{2}, \\ P(1) &= \frac{|1-e^{i\phi}|^2}{4} = \sin^2 \frac{\phi}{2} = \frac{1-\cos \phi}{2}. \end{aligned}$$

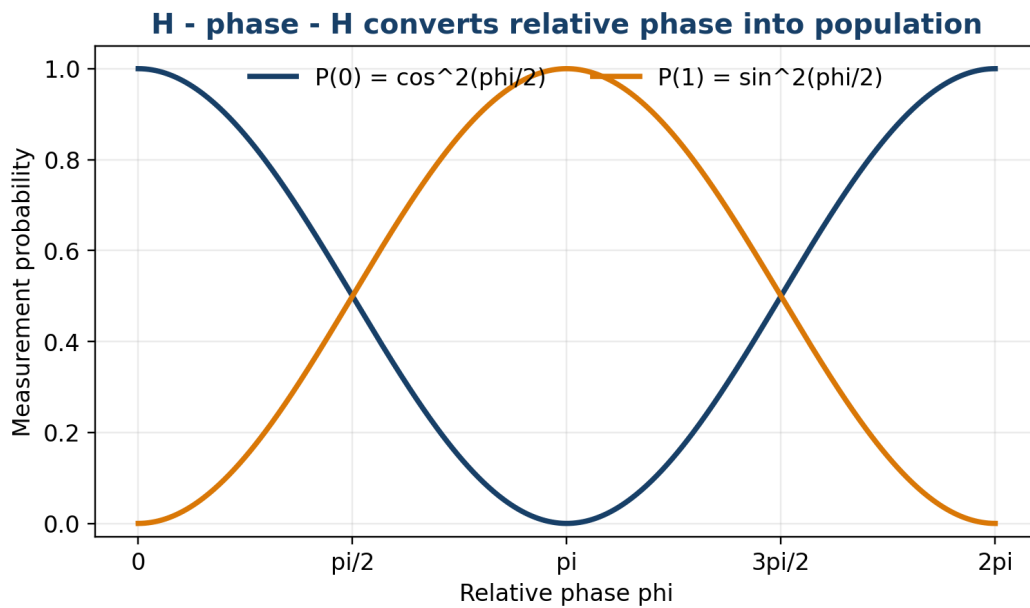


Figure 4: Interference fringes produced by converting a relative phase into computational-basis probabilities.

A single Z measurement cannot directly read the phase between $|0\rangle$ and $|1\rangle$. The final Hadamard supplies a phase reference and makes the two components overlap in the same output alternatives. Their relative phase then determines whether the overlap is constructive or destructive.

6. Constructive and destructive interference

Two complex amplitudes with equal magnitude reinforce when their phases agree and cancel when their phases differ by π :

$$\begin{aligned} a + a &= 2a && \text{constructive,} \\ a - a &= 0 && \text{destructive.} \end{aligned}$$

For general phases $a = re^{i\theta_1}$ and $b = re^{i\theta_2}$,

$$|a + b|^2 = 2r^2[1 + \cos(\theta_2 - \theta_1)].$$

Only the phase difference matters.

Interference is vector addition of complex probability amplitudes

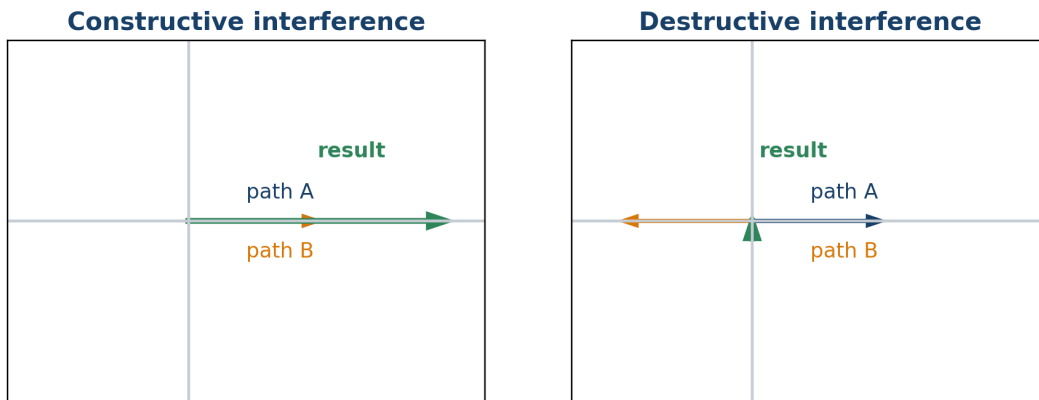


Figure 5: The same path magnitudes can produce maximal reinforcement or complete cancellation depending on relative phase.

6.1 Why amplitudes may cancel but probabilities cannot

Probabilities are nonnegative real numbers, so they cannot cancel. Amplitudes are complex vectors and can point in opposite directions. This is why a quantum circuit must preserve coherence until the desired recombination. Measuring too early replaces coherent amplitudes with classical alternatives and removes the phase-sensitive cross terms.

7. Two Hadamards: creation and recombination of paths

Since $H^2 = I$,

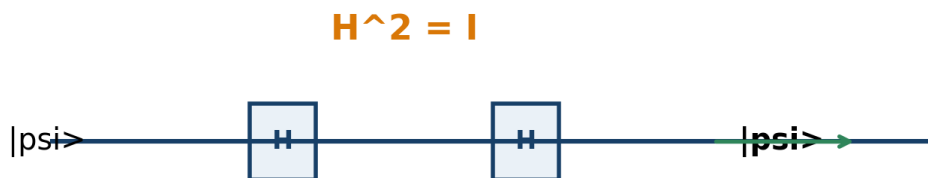
$$H(H|0\rangle) = H|+\rangle = |0\rangle, \quad H(H|1\rangle) = H|-\rangle = |1\rangle.$$

It is misleading to say that the first H creates randomness and the second somehow removes it. Before measurement, no random classical choice has occurred. The first H creates two coherent components; the second recombines them.

For input $|0\rangle$,

$$|0\rangle \xrightarrow{H} \frac{|0\rangle + |1\rangle}{\sqrt{2}} \xrightarrow{H} \frac{(|0\rangle + |1\rangle) + (|0\rangle - |1\rangle)}{2} = |0\rangle.$$

At output $|0\rangle$, the two path amplitudes add. At output $|1\rangle$, they cancel.



The first H creates two paths; the second H recombines them.

Figure 6: A double-Hadamard circuit is an elementary interferometer: split, preserve phase, and recombine.

If the qubit is measured after the first H , the state collapses to $|0\rangle$ or $|1\rangle$. A later H then produces a 50-50 result. The deterministic return to the input requires coherence between the two branches.

8. The elementary phase interferometer

The most important one-qubit interference circuit has the form

$$|0\rangle \xrightarrow{H} |+\rangle \xrightarrow{\text{phase}} \frac{|0\rangle + e^{i\phi}|1\rangle}{\sqrt{2}} \xrightarrow{H} \text{measure}.$$

Using $P(\phi) = \text{diag}(1, e^{i\phi})$,

$$HP(\phi)H|0\rangle = \frac{1 + e^{i\phi}}{2}|0\rangle + \frac{1 - e^{i\phi}}{2}|1\rangle.$$

This produces the fringe probabilities derived earlier.

Using the symmetric rotation $R_z(\phi) = \text{diag}(e^{-i\phi/2}, e^{i\phi/2})$,

$$HR_z(\phi)H = R_x(\phi).$$

Therefore the same interferometer may also be viewed as a rotation about the Bloch x axis.

ϕ	Intermediate state	$P(0)$	$P(1)$
0	$ +\rangle$	1	0
$\pi/2$	$(0\rangle + i 1\rangle)/\sqrt{2}$	1/2	1/2
π	$ -\rangle$	0	1
2π	$ +\rangle$	1	0

9. Optical interferometer analogy

A balanced beam splitter in a two-path optical interferometer plays a role analogous to a Hadamard transform. The first beam splitter creates a coherent superposition of paths. A relative phase accumulates. The second beam splitter mixes the paths again, and detector probabilities depend on that phase.

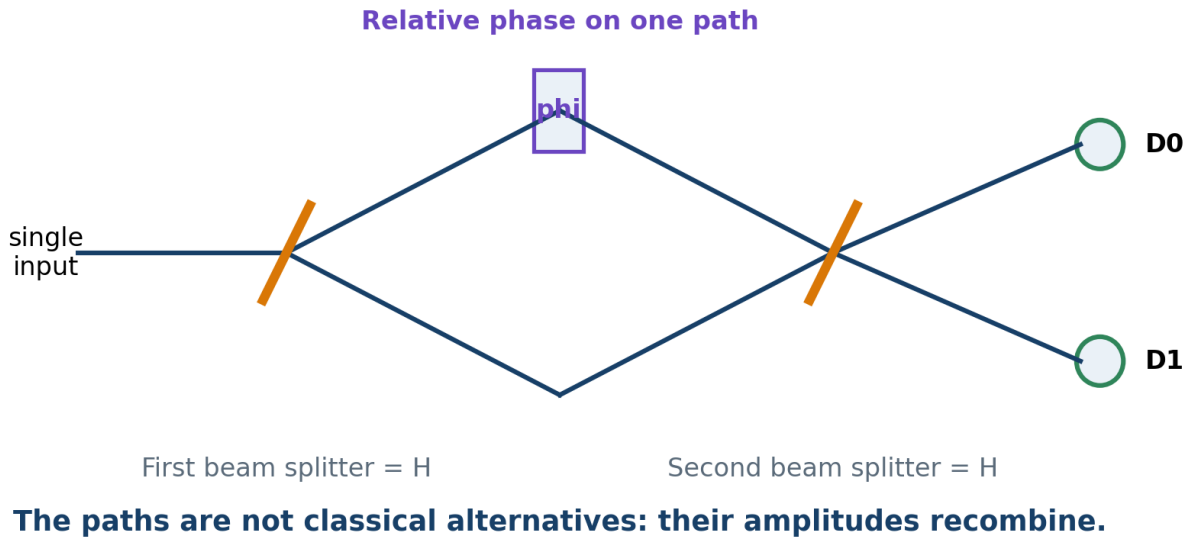


Figure 7: Conceptual Mach-Zehnder interferometer and its correspondence to an H -phase- H circuit.

The analogy is powerful but must be used carefully:

- the two paths can represent spatial modes, energy levels, polarization modes, or abstract computational states;
- a physical optical beam splitter has reflection phase conventions that may differ from the real-valued Hadamard matrix;
- global phases and port-label conventions can change matrices without changing observable detector probabilities.

If the environment records which path was taken, the alternatives become distinguishable. Their amplitudes no longer interfere fully. Interference is not destroyed by ignorance alone; it is destroyed when path information exists in principle in correlations with another system.

10. Ramsey interferometry

Ramsey interferometry is a standard method for estimating phase, frequency, and detuning in atomic, spin, superconducting, and other qubit platforms. A simplified sequence is

$$|0\rangle \xrightarrow{H} |+\rangle \xrightarrow{R_z(\phi)} \xrightarrow{H} \text{measure}.$$

During the free-evolution interval, a Hamiltonian proportional to Z produces

$$\phi = \Delta\omega t,$$

where $\Delta\omega$ is the detuning between the qubit transition and a reference oscillator. Repeating the experiment for different times produces oscillating populations called Ramsey fringes.

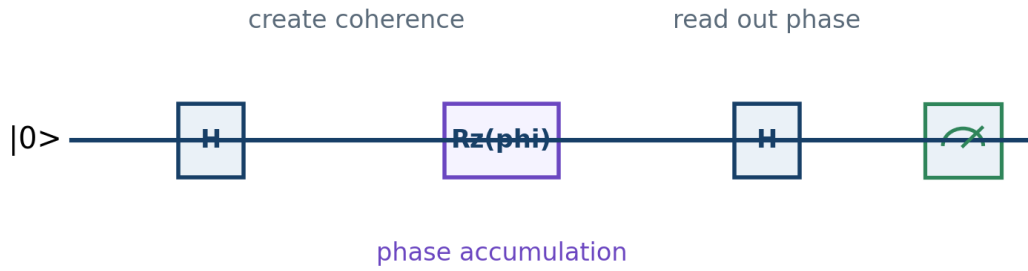


Figure 8: The two Hadamard-like pulses act as preparation and readout operations; the middle interval stores the unknown phase.

In a realistic experiment, coherence decays approximately as e^{-t/T_2^*} , leading to

$$P(0; t) = \frac{1}{2} \left[1 + e^{-t/T_2^*} \cos(\Delta\omega t + \phi_0) \right].$$

The oscillation frequency estimates detuning, while the envelope estimates dephasing.

11. Hadamard conjugation: exchanging bit and phase

Conjugating an operator A by H means applying HAH . The fundamental identities are

$$HXH = Z, \quad HZH = X, \quad HYH = -Y.$$

Hadamard swaps bit and phase operations

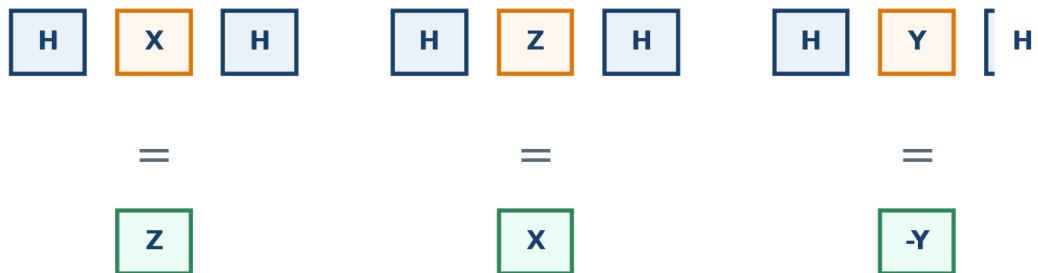


Figure 9: Conjugation by Hadamard exchanges X and Z , while reversing the sign of Y .

11.1 Derivation of $HZH = X$

$$\begin{aligned} HZH &= \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = X. \end{aligned}$$

11.2 Conceptual significance

Z writes a relative sign between computational components. Surrounding it with Hadamards converts that sign operation into a bit flip. Conversely, a bit flip in the computational basis becomes a phase flip in the diagonal basis. This duality appears throughout quantum error correction, stabilizer circuits, QKD, and algorithm design.

More generally,

$$HR_z(\theta)H = R_x(\theta), \quad HR_x(\theta)H = R_z(\theta).$$

12. Bloch-sphere action

A qubit density matrix can be written as

$$\rho = \frac{1}{2}(I + xX + yY + zZ).$$

After H ,

$$\rho' = H\rho H = \frac{1}{2}(I + zX - yY + xZ).$$

Therefore the Bloch vector transforms as

$$(x, y, z) \mapsto (z, -y, x).$$

The north pole $|0\rangle$ maps to the $+x$ direction $|+\rangle$, the south pole $|1\rangle$ maps to $-x$, and the y axis reverses.

The expectation values transform correspondingly:

$$\langle X \rangle_{H\psi} = \langle Z \rangle_{\psi}, \quad \langle Z \rangle_{H\psi} = \langle X \rangle_{\psi}, \quad \langle Y \rangle_{H\psi} = -\langle Y \rangle_{\psi}.$$

This is the observable form of basis exchange.

13. Interference and density matrices

For

$$\rho = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix},$$

apply H and measure in the computational basis. The probability of result 0 is

$$P(0) = \langle 0 | H\rho H | 0 \rangle = \frac{1}{2}(\rho_{00} + \rho_{11} + \rho_{01} + \rho_{10}).$$

Since $\rho_{00} + \rho_{11} = 1$ and $\rho_{10} = \rho_{01}^*$,

$$P(0) = \frac{1}{2} + \text{Re}(\rho_{01}), \quad P(1) = \frac{1}{2} - \text{Re}(\rho_{01}).$$

Thus the fringe contrast directly measures an off-diagonal density-matrix element.

To access its imaginary part, rotate the phase reference. Measuring Y can be implemented by applying $S^\dagger H$ before computational measurement, giving

$$\langle Y \rangle = -2\Im(\rho_{01})$$

under the standard Pauli convention.

Coherence is basis-dependent

Off-diagonal elements are not absolute objects; they depend on the chosen basis. A state diagonal in the X basis may have off-diagonal elements in the Z basis. Interference tests coherence between the alternatives that the interferometer actually recombines.

14. Multi-qubit Hadamards and uniform superposition

Applying H independently to n qubits gives

$$H^{\otimes n} |0^n\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle.$$

This is a uniform superposition over all n -bit strings.

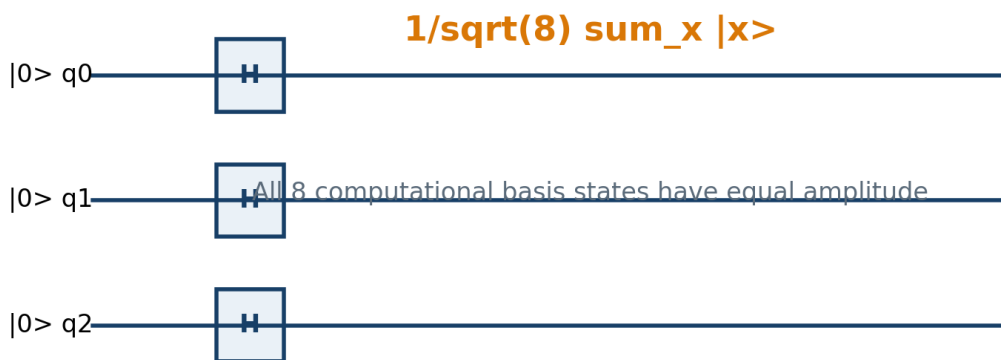


Figure 10: Three parallel Hadamard gates create equal amplitudes for all eight computational basis states.

The state does *not* mean that all 2^n answers can be read out. A measurement yields only one string. The useful resource is the possibility of transforming all amplitudes coherently before measurement.

14.1 Action on an arbitrary basis string

For $x, y \in \{0, 1\}^n$, define the mod-2 inner product

$$x \cdot y = x_1 y_1 \oplus x_2 y_2 \oplus \cdots \oplus x_n y_n.$$

Then

$$H^{\otimes n} |x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y \in \{0,1\}^n} (-1)^{x \cdot y} |y\rangle.$$

Every output has equal magnitude, but its sign depends on the binary inner product.

15. The Walsh-Hadamard transform

For an amplitude vector $\mathbf{a} = (a_x)$, the transformed amplitudes are

$$\tilde{a}_y = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{x \cdot y} a_x.$$

This is the normalized Walsh-Hadamard transform, the Fourier transform over the group $\mathbb{Z}_2^n$.

For two qubits,

$$H^{\otimes 2} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

Two-qubit Walsh-Hadamard transform H tensor H

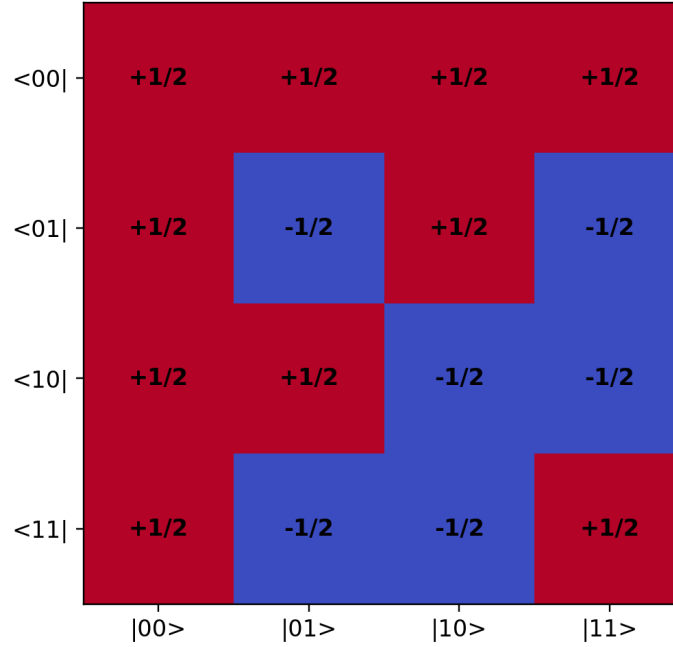


Figure 11: The signs of $H^{\otimes 2}$ encode parity relations between input and output bit strings.

15.1 Fast transform structure

A direct matrix-vector multiplication costs $O(n2^n)$ arithmetic operations if implemented through the recursive butterfly structure, rather than $O(4^n)$ for a generic dense matrix. A quantum circuit implements the transform with exactly n one-qubit Hadamard gates acting in parallel. This does not give classical access to all transformed amplitudes; it prepares the transformed quantum state.

16. Phase kickback and phase oracles

A Boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}$ is often represented by a reversible oracle

$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle.$$

Prepare the target in

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

Since $X|-\rangle = -|-\rangle$,

$$U_f |x\rangle |-\rangle = (-1)^{f(x)} |x\rangle |-\rangle.$$

The function value has been transferred to a phase on $|x\rangle$. This is **phase kickback**.

If the input register is uniform,

$$\frac{1}{\sqrt{2^n}} \sum_x |x\rangle \longrightarrow \frac{1}{\sqrt{2^n}} \sum_x (-1)^{f(x)} |x\rangle.$$

The computational probabilities are still uniform. The information resides in signs. A later Hadamard transform recombines the signed amplitudes and reveals correlations in f .

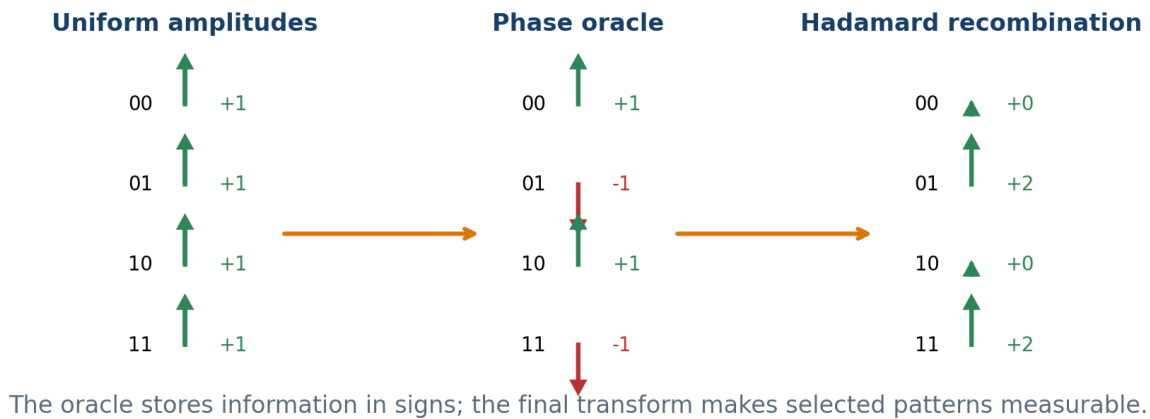


Figure 12: A phase oracle changes signs without changing immediate populations; the final Hadamard transform converts sign patterns into measurable amplitude patterns.

17. Deutsch's algorithm: the smallest complete example

Deutsch's problem promises that a one-bit Boolean function is either constant or balanced. Classically, one must evaluate both inputs in the worst case. The quantum algorithm uses one coherent oracle query.

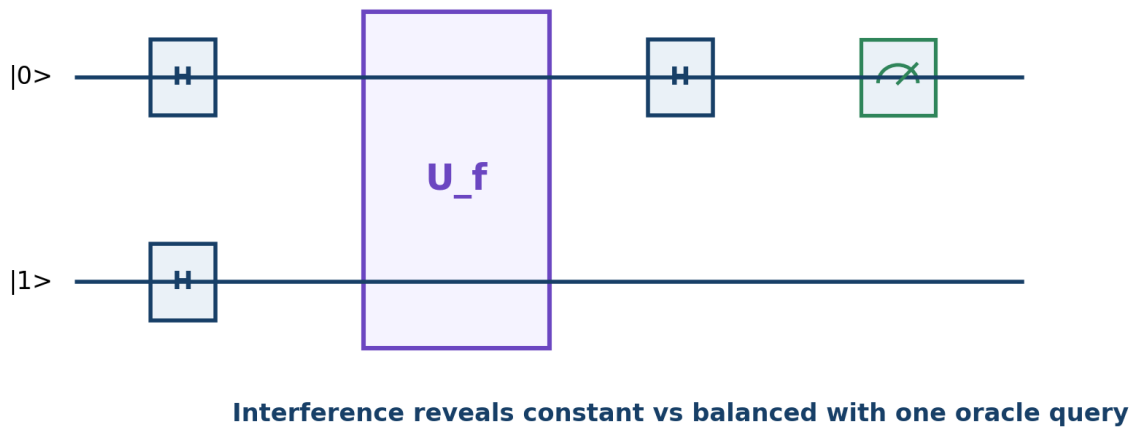


Figure 13: Deutsch's circuit uses phase kickback and Hadamard interference to identify a global property of the function.

After preparation,

$$|0\rangle |1\rangle \xrightarrow{H \otimes H} |+\rangle |-\rangle.$$

The oracle gives

$$\frac{(-1)^{f(0)} |0\rangle + (-1)^{f(1)} |1\rangle}{\sqrt{2}} \otimes |-\rangle.$$

Apply H to the first qubit. The amplitude of $|0\rangle$ is proportional to

$$(-1)^{f(0)} + (-1)^{f(1)},$$

and the amplitude of $|1\rangle$ is proportional to

$$(-1)^{f(0)} - (-1)^{f(1)}.$$

If $f(0) = f(1)$, the difference cancels and the result is 0. If they differ, the sum cancels and the result is 1.

The oracle does not write “constant” or “balanced” into a register. It writes two phases. The final Hadamard compares those phases by addition and subtraction. The answer is created by interference.

18. Interference patterns in larger algorithms

18.1 Bernstein-Vazirani

For a hidden string s , the oracle phase is $(-1)^{s \cdot x}$. Applying $H^{\otimes n}$ maps this entire phase pattern to the basis state $|s\rangle$. Each output bit is recovered through parity-sensitive interference.

18.2 Simon and period finding

Hadamards create and analyze superpositions over binary strings. In Simon’s algorithm, measurement after the final transform returns strings y satisfying $s \cdot y = 0$. Interference removes strings inconsistent with the hidden period.

18.3 Grover amplitude amplification

Grover’s algorithm alternates a phase reflection about marked states with a reflection about the uniform state. Hadamards implement the latter as

$$2|s\rangle\langle s| - I = H^{\otimes n}(2|0^n\rangle\langle 0^n| - I)H^{\otimes n},$$

where $|s\rangle = H^{\otimes n}|0^n\rangle$. Repeated reflections rotate amplitude toward the marked subspace.

18.4 Quantum phase estimation

Hadamards prepare control qubits in superposition. Controlled powers of a unitary imprint phases. An inverse quantum Fourier transform then converts phase relationships into a binary estimate. The final transform is more general than a tensor product of Hadamards, but the same principle holds: phase becomes population through interference.

18.5 Error correction and stabilizers

Hadamards exchange X -type and Z -type errors and stabilizers. For example, applying H to every qubit maps a code protecting against bit flips into the analogous phase-flip code description.

19. Hadamard gates in BB84 and QKD

BB84 uses two mutually unbiased bases:

$$Z : \{|0\rangle, |1\rangle\}, \quad X : \{|+\rangle, |-\rangle\}.$$

Hadamard gates convert between them:

$$H|0\rangle = |+\rangle, \quad H|1\rangle = |-\rangle, \quad H|+\rangle = |0\rangle, \quad H|-\rangle = |1\rangle.$$

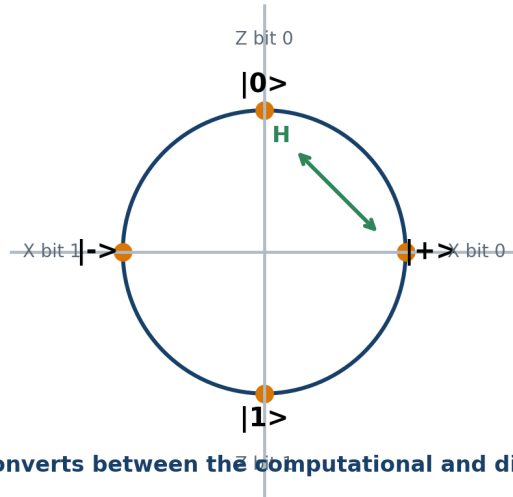


Figure 14: The Hadamard gate is the basis switch connecting the two BB84 encoding and measurement bases.

If a receiver measures in the correct basis, the bit is recovered deterministically in the ideal case. If the wrong basis is chosen, each outcome occurs with probability $1/2$ because

$$|\langle 0|+\rangle|^2 = |\langle 1|+\rangle|^2 = \frac{1}{2}.$$

An intercept-resend eavesdropper who guesses the basis incorrectly disturbs the state. After basis sifting, this creates detectable errors in a fraction of the retained data.

The Hadamard relation between mutually unbiased bases supplies measurement disturbance, but full QKD security also requires authenticated classical communication, parameter estimation, error correction, privacy amplification, finite-key analysis, and a clear device model.

20. Decoherence, distinguishability, and visibility

Suppose a balanced superposition has coherence reduced by a complex factor γ :

$$\rho = \frac{1}{2} \begin{pmatrix} 1 & \gamma e^{-i\phi} \\ \gamma^* e^{i\phi} & 1 \end{pmatrix}, \quad |\gamma| \leq 1.$$

After a Hadamard,

$$P(0) = \frac{1}{2} [1 + \text{Re}(\gamma e^{-i\phi})].$$

For real positive γ ,

$$P(0) = \frac{1}{2}(1 + \gamma \cos \phi).$$

The fringe visibility

$$V = \frac{P_{\max} - P_{\min}}{P_{\max} + P_{\min}}$$

equals $|\gamma|$ for this balanced model.

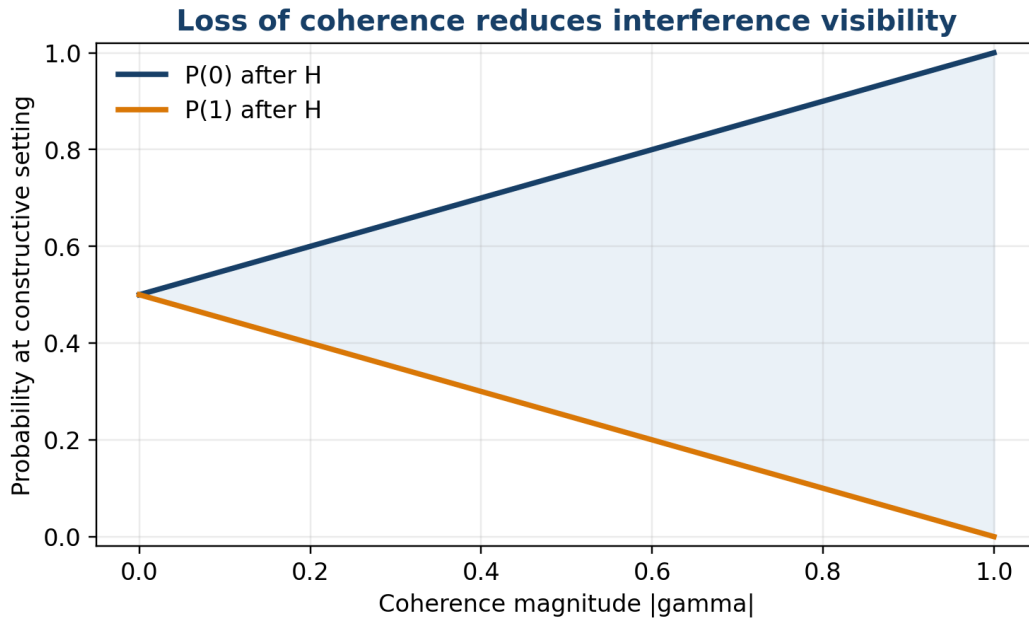


Figure 15: As coherence decreases, constructive and destructive outputs become less distinguishable.

20.1 Entanglement with an environment

Let path states correlate with environment states:

$$|\Psi\rangle = \frac{|0\rangle |E_0\rangle + e^{i\phi} |1\rangle |E_1\rangle}{\sqrt{2}}.$$

Ignoring the environment leaves qubit coherence proportional to

$$\gamma = \langle E_0 | E_1 \rangle.$$

If $|E_0\rangle = |E_1\rangle$, no path information exists and $|\gamma| = 1$. If they are orthogonal, the environment perfectly distinguishes the paths and $\gamma = 0$, eliminating the fringe.

A common complementarity relation is

$$V^2 + D^2 \leq 1,$$

where D quantifies path distinguishability.

21. Qiskit experiments

The following examples use the modern circuit and quantum-information interfaces. The core operations are *QuantumCircuit.h* for adding a Hadamard and *Statevector.from_instruction* for exact ideal-state analysis.

21.1 Verify the four basis transformations

```
from qiskit import QuantumCircuit
from qiskit.quantum_info import Statevector

for initial in [0, 1]:
    qc = QuantumCircuit(1)
    if initial == 1:
        qc.x(0)
    qc.h(0)
    state = Statevector.from_instruction(qc)
    print(initial, state)
```

Expected outputs are $|+\rangle$ and $|-\rangle$, up to display conventions and negligible floating-point error.

21.2 Build a phase interferometer

```
import numpy as np
from qiskit import QuantumCircuit
from qiskit.quantum_info import Statevector

phi = np.pi / 3
qc = QuantumCircuit(1)
qc.h(0)
qc.rz(phi, 0)
qc.h(0)

state = Statevector.from_instruction(qc)
print(state.probabilities_dict())
# Expected: P(0)=cos(phi/2)^2, P(1)=sin(phi/2)^2
```

For $\phi = \pi/3$, the ideal probabilities are 3/4 and 1/4.

21.3 Sample the fringe with measurements

```
import numpy as np
from qiskit import QuantumCircuit

circuits = []
for phi in np.linspace(0, 2*np.pi, 21):
    qc = QuantumCircuit(1, 1)
    qc.h(0)
    qc.rz(phi, 0)
    qc.h(0)
    qc.measure(0, 0)
    circuits.append(qc)

# Run these circuits with a local simulator or an IBM backend.
# Estimate P(0) from counts and compare it with cos(phi/2)**2.
```

Expected Qiskit measurement distributions for H-Rz(phi)-H

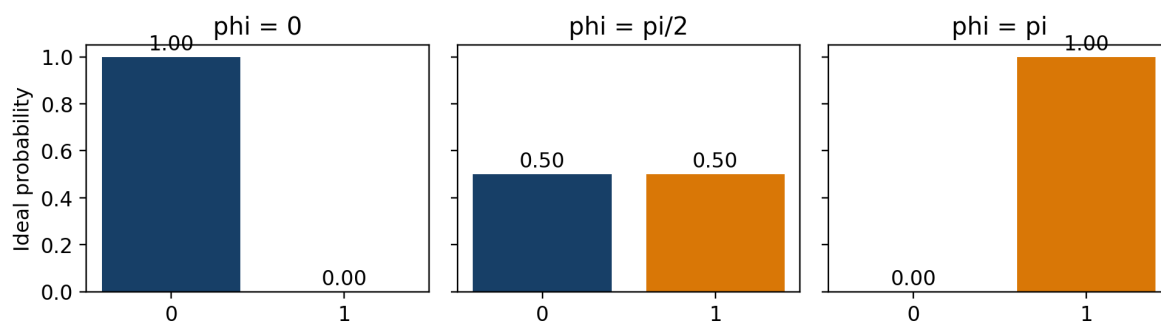


Figure 16: Ideal distributions at three characteristic phases. Finite-shot data fluctuate around these values.

21.4 Measure in the X basis

```
from qiskit import QuantumCircuit

qc = QuantumCircuit(1, 1)
# Prepare any state here.
qc.h(0) # rotate X basis to Z basis
qc.measure(0, 0) # 0 means +, 1 means -
```

For multiple qubits, printed bit strings may place the highest-index classical bit on the left. Always verify the mapping between qubits, classical bits, and displayed strings before interpreting a histogram.

22. Worked examples

Let

$$|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + i\sqrt{\frac{2}{3}}|1\rangle.$$

Then

$$H|\psi\rangle = \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{3}} + i\sqrt{\frac{2}{3}}\right)|0\rangle + \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{3}} - i\sqrt{\frac{2}{3}}\right)|1\rangle.$$

Each magnitude squared is

$$\frac{1}{2}\left(\frac{1}{3} + \frac{2}{3}\right) = \frac{1}{2},$$

because the cross term has zero real part. The output is balanced even though the input populations were $1/3$ and $2/3$.

A state has density matrix

$$\rho = \begin{pmatrix} 0.6 & 0.2 - 0.1i \\ 0.2 + 0.1i & 0.4 \end{pmatrix}.$$

After H , the probability of computational outcome 0 is

$$P(0) = \frac{1}{2} + \operatorname{Re}(\rho_{01}) = 0.5 + 0.2 = 0.7.$$

Thus $\langle X \rangle = P(0) - P(1) = 0.4 = 2 \operatorname{Re}(\rho_{01})$.

Compare

$$\rho_{\text{coh}} = |+\rangle\langle+|, \quad \rho_{\text{mix}} = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{I}{2}.$$

Both have equal computational populations. After H ,

$$H\rho_{\text{coh}}H = |0\rangle\langle 0|, \quad H\rho_{\text{mix}}H = \frac{I}{2}.$$

The coherent state gives outcome 0 with certainty; the mixture remains random. Equal populations do not imply equal quantum states.

An ideal H -phase- H experiment gives $P(0) = 0.9$. Since

$$P(0) = \frac{1 + \cos \phi}{2},$$

we obtain $\cos \phi = 0.8$. Therefore

$$\phi = \pm \arccos(0.8) \pmod{2\pi}.$$

A single cosine measurement cannot distinguish the two signs. A second experiment with a $\pi/2$ phase shift, effectively measuring the sine quadrature, resolves the ambiguity.

Using the binary inner-product formula,

$$H^{\otimes 2}|10\rangle = \frac{1}{2} \sum_{y_1 y_2} (-1)^{1 \cdot y_1 + 0 \cdot y_2} |y_1 y_2\rangle.$$

Hence

$$H^{\otimes 2}|10\rangle = \frac{1}{2}(|00\rangle + |01\rangle - |10\rangle - |11\rangle).$$

The sign depends only on the first output bit.

Evaluate $HZH|0\rangle$. Since $H|0\rangle = |+\rangle$ and $Z|+\rangle = |-\rangle$,

$$HZH|0\rangle = H|-\rangle = |1\rangle.$$

Thus $HZH = X$. A phase flip hidden between basis changes becomes a bit flip.

23. Common misconceptions and diagnostic checks

Misconception	Correction
“ H makes a qubit half 0 and half 1.”	A pure state has complex amplitudes, not hidden classical alternatives. Whether interference is possible distinguishes a superposition from a mixture.
“A superposition performs all computations and measurement reveals all answers.”	A measurement returns limited classical data. Algorithms must use interference to concentrate information into observable outcomes.
“The minus sign in $ -\rangle$ has no effect because probabilities are squared.”	It has no effect in a direct Z measurement but changes later interference. Relative signs are observable after recombination.
“Two Hadamards cancel because the first measurement was random.”	No intermediate measurement is present. The cancellation is coherent and deterministic.
“Global and relative phase are both unobservable.”	Global phase is unobservable for an isolated state; relative phase controls interference.
“Any noise merely changes probabilities slightly.”	Dephasing can erase off-diagonal coherence while leaving computational populations unchanged, directly removing interference.
“The optical beam splitter is exactly the matrix H .”	Physical phase conventions differ. Equivalent matrices may vary by input/output phases and still predict the same observable interference.
“Uniform superposition gives exponential readable memory.”	It provides 2^n amplitudes that can be transformed coherently, not individually inspected without exponential resources.

Fast diagnostic sequence

For any interference circuit, ask:

1. What are the coherent alternatives?
2. Which operation writes their relative phases?
3. Where are the alternatives recombined?
4. Which output amplitudes are sums, and which are differences?
5. Has any measurement, environment, or ancilla stored which-path information?
6. What observable is actually estimated from the final counts?

24. Exercises

A. Core calculations

1. Compute $H(\sqrt{3}|0\rangle + |1\rangle)/2$ and the two measurement probabilities.
2. Show directly that $H^\dagger H = I$ and $H^2 = I$.
3. Find $H|+i\rangle$ and $H|-i\rangle$, where $|\pm i\rangle = (|0\rangle \pm i|1\rangle)/\sqrt{2}$.
4. For $|\psi\rangle = (|0\rangle + e^{i\phi}|1\rangle)/\sqrt{2}$, derive $\langle X \rangle$, $\langle Y \rangle$, and $\langle Z \rangle$.
5. Prove that $P_H(0) - P_H(1) = \langle X \rangle$ for any qubit state, where P_H means apply H and then measure Z .
6. Determine all pure states that give $P_H(0) = P_H(1) = 1/2$.

B. Interference and phase

3. Starting from $|0\rangle$, calculate the final state of $HP(\phi)H$ including its global phase relation to an x rotation.
4. A phase interferometer reports $P(1) = 0.36$. Find the possible phases modulo 2π .
5. Insert a Z gate between two Hadamards. Predict the final result for inputs $|0\rangle$, $|1\rangle$, $|+\rangle$, and $|-\rangle$.
6. Suppose an intermediate projective Z measurement is performed after the first H but its result is discarded. Use density matrices to show that the final measurement is uniform.
7. A dephasing channel maps $\rho_{01} \mapsto \lambda\rho_{01}$. Derive the fringe visibility for an initially balanced state.
8. For environment overlap $\langle E_0|E_1\rangle = re^{i\delta}$, derive the shifted fringe $P(0; \phi)$.

C. Multi-qubit transforms and algorithms

3. Expand $H^{\otimes 3}|101\rangle$ explicitly.
4. Prove $(H^{\otimes n})^2 = I^{\otimes n}$.
5. Show that $H^{\otimes n} \left(2^{-n/2} \sum_x (-1)^{s \cdot x} |x\rangle \right) = |s\rangle$.
6. In Deutsch's algorithm, calculate the first-qubit state for all four one-bit Boolean functions.
7. For $f(x_1, x_2) = x_1 \oplus x_2$, compute the state after the phase oracle and after the final $H^{\otimes 2}$.
8. Explain why measuring immediately after a phase oracle generally reveals no information about $f(x)$.
9. Derive $H^{\otimes n} Z^{\otimes n} H^{\otimes n} = X^{\otimes n}$.
10. Design a circuit that measures the operator $X \otimes Z$ using only one-qubit basis changes and computational-basis measurements.

D. Conceptual and implementation tasks

3. Explain, without using the word "random," why $HH|0\rangle = |0\rangle$.
4. Compare the final state of a coherent $|+\rangle$ input with the mixed state $I/2$ after H .

5. Write Qiskit code to sample 41 phases from 0 to 2π and plot the measured fringe.
6. Modify the experiment to estimate both sine and cosine quadratures of an unknown phase.
7. Describe how the Hadamard gate is used by Alice and Bob in BB84 preparation and measurement.
8. Explain why an eavesdropper cannot measure an unknown BB84 state in both bases without disturbance.

25. Selected solutions

1. Core state calculation

$$|\psi\rangle = \frac{\sqrt{3}}{2} |0\rangle + \frac{1}{2} |1\rangle.$$

Therefore

$$H|\psi\rangle = \frac{\sqrt{3}+1}{2\sqrt{2}} |0\rangle + \frac{\sqrt{3}-1}{2\sqrt{2}} |1\rangle.$$

The probabilities are

$$P(0) = \frac{(\sqrt{3}+1)^2}{8} = \frac{2+\sqrt{3}}{4}, \quad P(1) = \frac{2-\sqrt{3}}{4}.$$

3. Hadamard on Y -basis states

$$H|+\mathbf{i}\rangle = \frac{1}{2}[(1+\mathbf{i})|0\rangle + (1-\mathbf{i})|1\rangle].$$

Factoring $e^{i\pi/4}$ gives

$$H|+\mathbf{i}\rangle = e^{i\pi/4} \frac{|0\rangle - \mathbf{i}|1\rangle}{\sqrt{2}} = e^{i\pi/4} |-\mathbf{i}\rangle.$$

Similarly,

$$H|-\mathbf{i}\rangle = e^{-i\pi/4} |+\mathbf{i}\rangle.$$

This is consistent with the Bloch transformation $y \mapsto -y$.

4. Bloch expectations of an equatorial state

For $|\psi\rangle = (|0\rangle + e^{i\phi}|1\rangle)/\sqrt{2}$,

$$\langle X \rangle = \cos \phi, \quad \langle Y \rangle = \sin \phi, \quad \langle Z \rangle = 0.$$

6. States producing balanced Hadamard outcomes

For a pure state with Bloch vector (x, y, z) , applying H and measuring Z gives $P(0) = (1+x)/2$. Balanced outcomes require $x = 0$. These are all pure states on the great circle in the yz plane. Equivalently, $\operatorname{Re}(\alpha^*\beta) = 0$.

7. Relation between phase gate and x rotation

Because

$$P(\phi) = e^{i\phi/2} R_z(\phi),$$

we obtain

$$HP(\phi)H = e^{i\phi/2} R_x(\phi).$$

The prefactor is global. Acting on $|0\rangle$,

$$HP(\phi)H|0\rangle = e^{i\phi/2} \left(\cos \frac{\phi}{2} |0\rangle - \mathbf{i} \sin \frac{\phi}{2} |1\rangle \right).$$

8. Infer phase from $P(1)$

$$P(1) = \sin^2(\phi/2) = 0.36.$$

Thus $|\sin(\phi/2)| = 0.6$. The solutions modulo 2π are

$$\phi = 2 \arcsin(0.6), \quad \phi = 2\pi - 2 \arcsin(0.6).$$

10. Destroying interference by an unread measurement

After the first H ,

$$\rho = |+\rangle\langle+| = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

An unread computational measurement removes the off-diagonal terms:

$$\rho \longrightarrow \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{I}{2}.$$

Since $HIH = I$, the final state remains $I/2$, so both outputs have probability $1/2$.

12. Environment-induced fringe shift

Tracing out the environment gives coherence $re^{i\delta}e^{-i\phi}/2$. Therefore

$$P(0; \phi) = \frac{1}{2}[1 + r \cos(\phi - \delta)].$$

The magnitude r controls visibility and δ shifts the fringe.

13. Three-qubit expansion

For $x = 101$ and $y = y_1y_2y_3$, the sign is $(-1)^{y_1+y_3}$. Hence

$$\begin{aligned} H^{\otimes 3} |101\rangle &= \frac{1}{\sqrt{8}} (|000\rangle - |001\rangle + |010\rangle - |011\rangle \\ &\quad - |100\rangle + |101\rangle - |110\rangle + |111\rangle). \end{aligned}$$

15. Recovering a linear phase pattern

Start with

$$|\psi_s\rangle = 2^{-n/2} \sum_x (-1)^{s \cdot x} |x\rangle.$$

The amplitude of output y after $H^{\otimes n}$ is

$$2^{-n} \sum_x (-1)^{s \cdot x + x \cdot y} = 2^{-n} \sum_x (-1)^{x \cdot (s \oplus y)}.$$

The sum equals 1 when $y = s$ and 0 otherwise, so the output is $|s\rangle$.

17. Parity function phase pattern

For $f(x_1, x_2) = x_1 \oplus x_2$, the signed uniform state is

$$\frac{1}{2}(|00\rangle - |01\rangle - |10\rangle + |11\rangle).$$

This equals $|-\rangle \otimes |-\rangle$. Applying $H \otimes H$ gives $|11\rangle$.

20. Measuring $X \otimes Z$

Apply H to the first qubit only, then measure both in the computational basis. Interpret outcome bits as eigenvalues $(-1)^{b_1}$ and $(-1)^{b_2}$. Their product $(-1)^{b_1+b_2}$ is one sample of the $X \otimes Z$ observable.

23. Estimating both quadratures

The standard H readout estimates $\cos \phi$. To estimate $\sin \phi$, shift the analysis phase by $\pi/2$, for example by applying $S^\dagger$ before the final H . Repeating both circuits provides estimates of $(\cos \phi, \sin \phi)$ and therefore $\phi = \text{atan2}(\sin \phi, \cos \phi)$.

26. Compact reference sheet

Core identities

$$\begin{aligned}
 H &= \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, & H^\dagger &= H, & H^2 &= I, \\
 H|0\rangle &= |+\rangle, & H|1\rangle &= |-\rangle, & H|+\rangle &= |0\rangle, & H|-\rangle &= |1\rangle, \\
 H(\alpha|0\rangle + \beta|1\rangle) &= \frac{\alpha + \beta}{\sqrt{2}}|0\rangle + \frac{\alpha - \beta}{\sqrt{2}}|1\rangle, \\
 HXH &= Z, & HZH &= X, & HYH &= -Y, \\
 H R_z(\phi) H &= R_x(\phi), & H R_x(\phi) H &= R_z(\phi).
 \end{aligned}$$

Interference formulas

For $|\psi(\phi)\rangle = (|0\rangle + e^{i\phi}|1\rangle)/\sqrt{2}$,

$$P_H(0) = \cos^2(\phi/2) = \frac{1 + \cos \phi}{2}, \quad P_H(1) = \sin^2(\phi/2) = \frac{1 - \cos \phi}{2}.$$

For a general density matrix,

$$P_H(0) = \frac{1}{2} + \text{Re}(\rho_{01}), \quad \langle X \rangle = P_H(0) - P_H(1) = 2 \text{Re}(\rho_{01}).$$

Multi-qubit transform

$$\begin{aligned}
 H^{\otimes n} |x\rangle &= 2^{-n/2} \sum_y (-1)^{x \cdot y} |y\rangle, \\
 \tilde{a}_y &= 2^{-n/2} \sum_x (-1)^{x \cdot y} a_x.
 \end{aligned}$$

A phase oracle creates $(-1)^{f(x)}$ signs; a later transform converts their correlations into output probabilities.

Mental model

1. Hadamard creates or changes a coherent path basis.
2. Relative phase stores information between paths.
3. A later Hadamard adds and subtracts amplitudes.
4. Constructive interference amplifies selected outcomes.

5. Destructive interference suppresses others.
6. Decoherence removes the cross terms required for this process.

Where this leads next

The next topic is the Bloch sphere as a complete geometric language for pure one-qubit states, rotations, measurement axes, and mixed-state contraction. The present volume already used some Bloch-vector facts; the next volume develops them systematically and connects them to practical circuit design and tomography.

References and further study

- M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*.
- J. Preskill, *Lecture Notes on Quantum Computation*.
- IBM Quantum Documentation, current Qiskit *QuantumCircuit* and *Statevector* API references.
- Standard treatments of Ramsey spectroscopy, two-path interference, Walsh transforms, and BB84 quantum key distribution.